

DSAA3073: Theories in Data Science

Week 6 Validation Sheet · Honest Model Selection and Evaluation

Coverage: Week 6

Duration: 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. A team fits 20 declared candidates on training data, chooses the one with the smallest validation loss, refits it on training plus validation data, and wants a final performance estimate. Which statement is sound?

- (a) The minimum validation loss is final evidence because validation data were not used for fitting.
- (b) The training loss after refitting is final evidence because it uses more data.
- (c) A sealed test set may evaluate the completed selection-and-refitting procedure.
- (d) Averaging the training and validation minima removes selection optimism.

A2 [5 marks]. In LOOCV on a dataset of size n , what does each held-out loss directly evaluate?

- (a) A predictor trained on all n observations.
- (b) The declared algorithm trained on $n - 1$ observations.
- (c) A predictor trained on one observation.
- (d) A deterministic population quantity with no sampling variability.

A3 [5 marks]. Which setting licenses the exact residual correction $e_{-i,i} = \frac{e_i}{1-h_{\lambda,ii}}$ taught in Week 6?

- (a) Any algorithm whose full-data fitted values can be written as Hy .
- (b) Ordinary k -NN, even when deletion changes the neighbour set.
- (c) Fixed-design ordinary or ridge least squares with compatible deletion fits, unchanged features and penalty, and $1 - h_{\lambda,ii} \neq 0$.
- (d) Any fitted rule after its tuning parameter has been selected from the responses.

A4 [5 marks]. Which statement correctly distinguishes the Week 6 criteria?

- (a) AIC is a distribution-free finite-sample PAC bound.
- (b) BIC proves that its selected model is true on any finite candidate list.
- (c) AIC targets expected fresh conditional KL or log-score discrepancy under regular large-sample likelihood conditions.
- (d) AIC, BIC, CV, and SRM differ only by interchangeable complexity constants.

Part B: Definition in use

B1 [20 marks]. A public-health team has 800 i.i.d. observations and a small, predeclared collection of regular Poisson regression models fitted by maximum likelihood. Every free likelihood parameter is counted. Its scientific target is fresh conditional log-score performance. The team can seal 160 observations before any model comparison, and it wants both a selected model and a numerical final performance report.

Recommend a selection-and-evaluation workflow. Justify the selection rule from the target and candidate family, trace which data may influence each decision, and explain why neither a BIC identification claim nor an SRM PAC guarantee is obtained automatically.

Answer space

Part C: Basic skills

C1 [7 marks]. Three cross-validation folds have sizes 2, 3, 5 and mean held-out squared losses 1.0, 3.0, 2.0, respectively. Compute the observation-weighted CV score. Also compute the unweighted average of the three fold means and state why it is not the required score.

Answer space

C2 [8 marks]. A regular Gaussian regression fitted by maximum likelihood has three regression coefficients and an estimated noise variance. At its MLE, $\ell(\hat{\theta}) = -50$, with $n = 100$ and $\log 100 \approx 4.605$. Compute its standard AIC and BIC under the lower-is-better convention. Then state in one sentence what neither number is: a PAC risk certificate or an untouched final-performance estimate.

Answer space

Part D: Construction and design

D1 [17 marks]. Consider ordinary one-nearest-neighbour regression on

$$(x_1, x_2, x_3) = (0, 1, 3), \quad (y_1, y_2, y_3) = (0, 1, 4).$$

The full-data in-sample convention includes the query point itself, so the fitted-response map is $\hat{y} = I_3 y$.

Use this dataset to construct a concrete demonstration that the least-squares PRESS residual formula is not licensed merely by a full-data linear map. Compute the actual leave-one-out predictions and score, then identify exactly what fails in the attempted PRESS calculation.

Answer space

D2 [18 marks]. A team adaptively compares 30 tree-based classification pipelines, including feature transformations, depths, and probability thresholds. There is no declared regular likelihood parameter count. It proposes four outer folds and five-fold CV inside every outer-training portion.

Design one complete legal inner/outer pipeline and state what the final outer average estimates. Your design must place every adaptive choice, refit, and scored observation in the correct data role, and must rule out feedback from outer outcomes.

Answer space

Part E: Proof and derivation

E1 [25 marks]. A proposed regression procedure receives two candidate feature columns

$$z^{(1)} = (0, 0, 1)^T, \quad z^{(2)} = (0, 1, 1)^T$$

and a response vector $y(a) = (-a, -a, a)^T$, where $a > 0$. It first selects the column maximizing $|z^T y|$, with ties resolved in favour of $z^{(1)}$, and then fits one-dimensional ridge with $\lambda = 1$ on the selected column. An analyst takes the matrix and leverages from the full-data selected ridge fit and applies the PRESS correction to every case, claiming exact LOOCV for the entire procedure.

Give one coherent mathematical audit valid for every $a > 0$. Develop a held-out-prediction criterion, with the necessary quantifiers, for when frozen-full-selection PRESS and LOOCV obtained by rerunning the entire procedure agree; distinguish a useful sufficient selector condition from what is actually necessary. Use the displayed family to decide the analyst's claim, and then design an honest inner/outer evaluation of this adaptive procedure with a justification that no outer response can influence the prediction on which its own outer loss is computed.

Answer space

Part F: Synthesis and limitations

F1 [15 marks]. A colleague proposes the universal identity

$$\text{generalization score} = \text{training fit} + \text{complexity penalty}$$

and says CV, AIC, BIC, and SRM differ only in the penalty. Reject or defend the proposal by comparing, for all four methods, the object actually scored, the conditions that license it, and the conclusion it can support.

Answer space